

Silicon Analysis: An Inspectable Browser Tool for CMOS Power, Performance, Area, Yield, and Cost Exploration

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Abstract

Early technology selection requires reasoning about logic density, memory scaling, operating frequency, power, manufacturing yield, and cost together. Silicon Analysis implements this workflow in CMOS PPAC Scaling Studio, a self-contained HTML application with an embedded process library and no external JavaScript dependencies. Its 41 selectable entries cover six process families and combine publicly motivated scaling factors with explicit estimation and roadmap assumptions. The application separates logic, SRAM, and fixed area; evaluates power at both source and estimated target frequencies; combines a negative-binomial yield model with wafer, package/test, and amortized non-recurring costs; and provides a deterministic rectangular-die wafer map, including optional wafer-to-wafer stacking. We document the implemented equations and reproduce experiments directly from the application's JavaScript. For its default 400 mm², 10 W, 1 GHz TSMC 12 nm design, the TSMC 6 nm scenario produces 135.59 mm², 6.178 W at 1.20 GHz, 79.03% modeled yield, and \$36.07 per good die under the default cost assumptions. Sensitivity experiments expose the effects of fixed area, SRAM, wafer-cost basis, and stack yield. Arithmetic checks cover all 1,681 source–target combinations. These results establish reproducibility and illustrate model behavior; they do not establish prediction accuracy against measured silicon.

1 Introduction

Selecting a fabrication process changes more than transistor density. A logic-dominated design can shrink substantially while its memory macros and analog interfaces scale differently. A faster operating point can consume more dynamic power than an equal-frequency comparison suggests. Higher wafer cost and defect density can offset the benefit of a smaller die. A useful early estimator must therefore expose the assumptions connecting power, performance, area, and cost (PPAC), rather than reduce the decision to a nominal node label.

This paper describes the existing Silicon Analysis repository¹, whose interface is named CMOS PPAC Scaling Studio. The application accepts a reference design and a source/target process pair, then displays estimated area, power, frequency, yield, and cost. Its implementation is contained in one HTML file: the process library, model functions, interface, and Canvas rendering are inspectable together. A separate wafer-map view makes geometric edge losses and sampled die failures visible.

Our contribution is a documented, reproducible integration of these calculations. Specifically, we provide (i) a mathematical specification of the implemented PPAC, yield, and cost paths; (ii) an architecture description that distinguishes analytical estimates from the spatial wafer model; and (iii) code-derived experiments and a source-provenance audit. The work does not introduce a transistor model, a physical-design optimizer, or a new statistical yield law. It does

¹https://github.com/ldcyes/silicon_analysis

not report fabrication measurements, timing signoff, predictive error against a held-out dataset, or performance superiority over other tools.

2 Background and relationship to prior work

Stillmaker and Baas describe polynomial technology-scaling models derived from HSpice simulations using PTM and ITRS inputs [1]. Sarangi and Baas’s DeepScaleTool fits published silicon data to estimate area, delay, and energy scaling in the deep-submicron regime [2]. These works motivate source-to-target factor ratios. Silicon Analysis uses an explicit lookup table of normalized factors; it does not reproduce their fitting procedures or inherit their reported accuracy. In particular, this repository’s 2 nm-class and roadmap entries extend beyond the process range studied by DeepScaleTool.

Negative-binomial defect models are part of a broader family of compound statistical yield models. Bogdanov et al. discuss hierarchical fault distributions across regions, wafers, and batches [3]. The present implementation uses a single negative-binomial survival expression at each die location, with an optional radial adjustment. It does not model the complete hierarchical distribution or generate spatially correlated defect clusters.

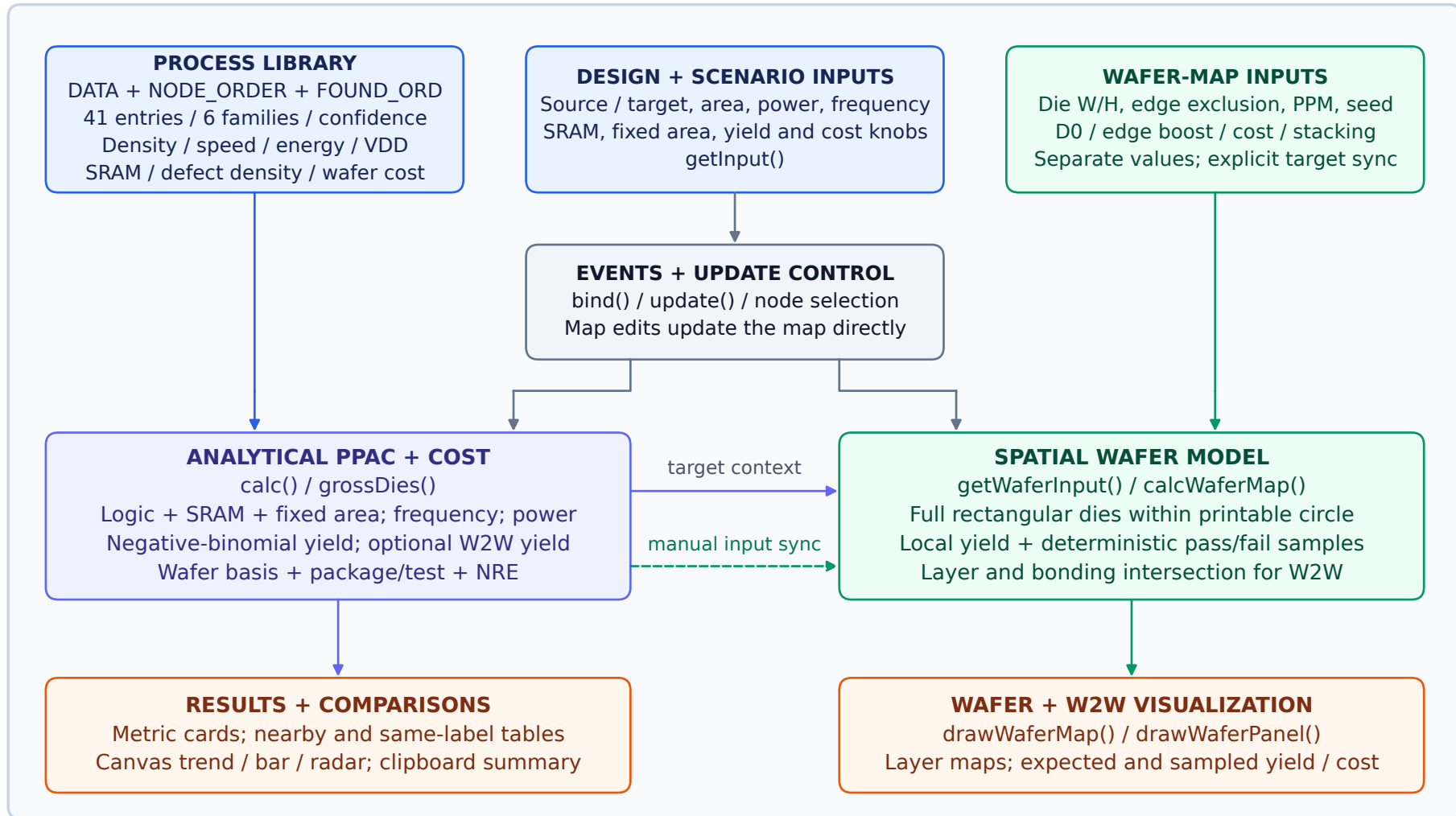
Public foundry announcements provide some useful relative anchors. TSMC’s 2019 N6 announcement states an 18% logic-density improvement over N7 [4]; the repository encodes this as $7.08/6.00 = 1.18$. GlobalFoundries’ 22FDX announcement describes FD-SOI and a 20% smaller die than 28 nm [5]; the corresponding density ratio in the library is 1.25, whose reciprocal is 0.8. These examples check the interpretation of two source statements. They do not validate every field in either record.

3 Implementation and process-library provenance

Figure 1 shows the application at repository revision `95fefca` (the full revision is recorded in the source ledger). There is no server-side model or persistent database. Input values are held in DOM controls, while the process library and display state are JavaScript objects. The page can be opened as a local file or served by any static HTTP server. Clipboard access depends on browser permissions. Reloading the page restores its defaults.

CMOS PPAC Scaling Studio

Architecture overview / one HTML file / local browser execution



Solid arrows: calculation / control flow. Dashed: Sync from Target Node. Local state resets on reload; no backend or external JS.

Figure 1: Implemented browser architecture. Solid arrows show calculation or control flow. The dashed arrow denotes explicit copying of selected target values into the controls consumed by the wafer model. Map controls retain their own values between synchronizations. The editable source is `docs/architecture.drawio`.

3.1 Model and presentation boundaries

`getInput()` reads the main controls and `calc()` evaluates a source–target pair. `update()` obtains the selected result and invokes the metric-card, wafer-map, table, and chart updates. Trend charts repeat `calc()` over the selected families and available nodes. `nearbyNodes()` selects at most two neighboring entries on either side in the foundry’s supported-node sequence. Same-label comparisons include only families containing that numeric label. The bar and radar plots normalize values within the displayed comparison set; their scores are visual aids, not absolute engineering objectives.

The map path uses `getWaferInput()` and `calcWaferMap()`. Editing map controls recalculates that view directly. Main-input changes refresh the map but do not automatically overwrite its width, height, defect-density, or wafer-cost controls. The *Sync from Target Node* button copies selected values, including square-die dimensions derived from target area. The dimensions are rounded to two decimals, D_0 to three, and wafer cost to whole dollars. Edge exclusion, radial boost, extra failures in parts per million (PPM), and random seed retain their current values. Consequently, a refreshed map and the analytical cards need not describe identical geometries or failure assumptions.

3.2 Coverage and confidence labels

Table 1 reports the selectable library. The global order is 28, 22, 16, 14, 12, 10, 7, 6, 5, 4, 3, and 2. The keys are comparison labels, not measured physical dimensions. Some records explicitly represent interpolated classes or grouped process families, such as Intel 14++/12-class and 18A/20A-class. GlobalFoundries 22FDX remains a distinct 22 nm FD-SOI record. Its missing advanced nodes are consistent with the company’s 2018 announcement placing its 7 nm program on hold [6]; library coverage should not be interpreted as a current foundry-service catalog.

Table 1: Selectable process entries at the evaluated revision. H, M, and R are the repository’s High, Medium, and Roadmap labels.

Family	Numeric labels	Count	H	M	R
TSMC	28,16,12,10,7,6,5,4,3,2	10	6	4	0
Samsung	28,14,12,10,7,6,5,4,3	9	5	4	0
Intel	14,12,10,7,4,3,2	7	5	2	0
SMIC	28,14,7	3	1	2	0
GF	28,22,14,12	4	3	1	0
Huawei τ	14,10,7,6,5,4,3,2	8	0	1	7
Total		41	20	14	7

The availability predicate excludes records labeled `low` or `alias`; it does not exclude every record whose descriptive text uses mapping or interpolation. High and Medium are qualitative labels attached to complete records, not field-level confidence intervals. A High record can still include estimated power, defect-density, or cost parameters.

The HTML attributes wafer-cost baselines to previously uploaded IBS-style tables and Huawei LogicFolding values to uploaded slides. Neither those tables nor those slides are present in the evaluated repository. Their titles, dates, and original contents therefore cannot be independently checked from this artifact. We retain these numbers as repository assumptions and do not cite a fabricated primary source for them. Similarly, the broad TSMC, Samsung, Intel, and SMIC source descriptions in the interface are not a field-by-field provenance record. The accompanying source ledger separates verified contextual references from unsupported parameter provenance.

4 Analytical model

Unless specified otherwise, inputs are nonnegative physical quantities, with strictly positive reference frequency, process factors, die area, and clustering parameter. Numerical floors in the implementation prevent some singular calculations; they do not replace complete input validation.

4.1 Area and performance

For process p , let d_p , s_p , and e_p be normalized logic-density, speed, and same-frequency dynamic-energy factors; let m_p be SRAM density in Mbit/mm² and v_p the nominal supply voltage. Subscripts s and t refer to source and target processes. Given source logic area $A_{\ell,s}$, SRAM capacity M , and fixed area A_o , the target estimate is

$$A_{\ell,t} = A_{\ell,s} \frac{d_s}{d_t} h_A, \quad A_{m,t} = \frac{M}{m_t}, \quad (1)$$

$$A_t = \max(10^{-4}, A_{\ell,t} + A_{m,t} + A_o), \quad f_t = f_s \frac{s_t}{s_s} h_f. \quad (2)$$

SRAM area is zero when capacity is zero. h_A is a logic-area overhead factor and does not multiply SRAM or fixed area. h_f is a user-selected frequency-realization factor representing unmodeled implementation effects. The term *converged frequency* in the interface means this estimated frequency; no timing-closure iteration is executed.

For most records the displayed standardized transistor density is $15.3d_p$ MTr/mm², using a repository constant for the 28 nm baseline. Some Huawei records instead contain explicit standardized density, raw density, and an effective density motivated by slide notes. Area uses `density`, not the displayed standardized-density field. The implemented 2/3 conversion of Huawei raw MTR is a repository convention, not a universal conversion between transistor-counting methods.

4.2 Logic and SRAM power

Let $P_{\ell,s}$ be the source logic-power input. Despite its *active power* label, the code assigns a fixed fraction $\lambda = 0.08$ to leakage and the remaining fraction to dynamic power. Define

$$r_f = f_t/f_s, \quad r_e = (e_t/e_s)h_P, \quad (3)$$

$$r_{\text{leak}} = \max \left[0.08, \left(\frac{A_{\ell,t}}{A_{\ell,s}} \right)^{0.72} \left(\frac{v_t}{v_s} \right)^{1.2} \right], \quad (4)$$

$$P_{\ell,\text{same}} = (1 - \lambda)P_{\ell,s}r_e + \lambda P_{\ell,s}r_{\text{leak}}, \quad (5)$$

$$P_{\ell,t} = (1 - \lambda)P_{\ell,s}r_e r_f + \lambda P_{\ell,s}r_{\text{leak}}. \quad (6)$$

The implementation floors numerator and denominator logic areas at 10^{-9} in Eq. 4. The exponents, leakage fraction, and minimum leakage scale are fixed heuristics; the repository contains no calibration dataset for them. The power guardband h_P acts on logic dynamic energy only. It does not directly scale leakage or SRAM power.

When both $M > 0$ and source SRAM dynamic power $P_{m,s} > 0$, the SRAM contribution is

$$P_{m,\text{same}} = P_{m,s} \frac{m_s}{m_t} \left(\frac{v_t}{v_s} \right)^2, \quad P_{m,t} = P_{m,\text{same}} r_f. \quad (7)$$

Otherwise it is zero. The displayed totals are $P_{\text{same}} = P_{\ell,\text{same}} + P_{m,\text{same}}$ and $P_t = P_{\ell,t} + P_{m,t}$. Fixed-area blocks contribute no additional power. The inverse-density SRAM power factor is an estimator heuristic, not a macro-level energy model.

4.3 Yield and cost

For wafer diameter D in mm, the analytical gross-die approximation is

$$G(A, D) = \max \left(0, \frac{\pi(D/2)^2}{A} - \frac{\pi D}{\sqrt{2A}} \right). \quad (8)$$

The code returns zero when $A \leq 0$. G can be fractional, and the unrounded value is used for cost. This approximation has no explicit die aspect ratio, scribe street, reticle limit, or user-specified edge exclusion.

Let $A_c = A_t/100$ convert mm^2 to cm^2 , D_0 be target defect density in cm^{-2} multiplied by a scenario factor m_D , and $\alpha > 0$ be the clustering parameter. With line/parametric yield y_{line} ,

$$Y_d = \left(1 + \frac{D_0 A_c}{\alpha} \right)^{-\alpha}, \quad q = Y_d y_{\text{line}}. \quad (9)$$

For a conventional record, final yield is q . For a stack-enabled Huawei record with $L \in \{2, 3\}$ layers and bonding yield b per interface,

$$Y_L = q^L b^{L-1}. \quad (10)$$

This assumes identical layer footprints and yields and independent layer and bonding failures. It does not include known-good-die selection or repair. Changing the layer selector changes yield and wafer multiplicity; it does not independently improve the record's area, frequency, or power factors.

Let W be the selected per-wafer baseline, m_C a wafer-cost multiplier, P_{pkg} package/test cost per good die, N_{NRE} non-recurring engineering expense, and V production volume. Then

$$C_{\text{Si}} = \frac{LWm_C}{G(A_t, D)Y_L}, \quad C_{\text{good}} = C_{\text{Si}} + P_{\text{pkg}} + \frac{N_{\text{NRE}}}{\max(1, V)}. \quad (11)$$

For conventional records set $L = 1$. Zero gross dies or zero final yield produce infinite silicon cost, displayed as an em dash in the interface. The default cost basis is the stored introductory wafer price (**intro**); the manufacturing alternative is **wafer**. Most families share these node-indexed baselines. They are scenario inputs, not foundry-specific quotes. NRE amortization interprets V as the count over which the user intends to distribute that expense, without an additional yield penalty.

5 Spatial wafer and stack visualization

For map die dimensions w and h , wafer radius $D/2$, and edge exclusion e_{edge} , the code sets printable radius $R = \max(1, D/2 - e_{\text{edge}})$. Die centers occupy a fixed lattice $((i + 1/2)w, (j + 1/2)h)$. A candidate is retained only if all four corners lie within the printable circle. Lattice translation and die orientation are not optimized.

At retained die i , let ρ_i be radial distance divided by R and β the edge-defect boost. Its local defect density and single-layer pass probability are

$$D_{0,i} = D_0(1 + \beta\rho_i^4), \quad (12)$$

$$q_i = \text{clip}_{[0,1]} \left[\left(1 + \frac{D_{0,i}wh}{100\alpha} \right)^{-\alpha} y_{\text{line}} \left(1 - \frac{\text{PPM}}{10^6} \right) \right]. \quad (13)$$

The map's expected final yield is

$$\bar{Y} = \frac{1}{N} \sum_{i=1}^N q_i^L b^{L-1}. \quad (14)$$

With radial heterogeneity and $L > 1$, this generally differs from $(N^{-1} \sum_i q_i)^L b^{L-1}$. The implementation sums location-specific stack probabilities as in Eq. 14. Its short UI annotation raising average layer yield to a power is therefore only an approximation for spatially varying yields.

Each layer and bonding interface uses a deterministic integer-hash draw indexed by lattice position, the user seed, and distinct seed offsets. A location passes if all its layer and interface draws pass. The sampled yield is the passing fraction; expected cost uses $N\bar{Y}$, while sampled cost uses the realized number of passing locations. Repeating inputs and seed repeats the map. Changing the seed changes the realization, not the expected probability. These drawings are synthetic outcomes, not measured wafer maps, and no spatial-correlation calibration of the hash is claimed.

The analytical model evaluates one process pair in constant time. A wafer map scans a lattice proportional to wafer area divided by die area, with $O(NL)$ work and stored per-die outcomes. Canvas drawing subsamples maps above approximately 12,000 retained dies per pane, but the model still enumerates the full lattice. The experiments below do not measure latency, and very small die dimensions can make synchronous browser updates costly.

6 Reproducible experiments

6.1 Protocol and checks

The accompanying `scripts/evaluate.mjs` reads the original HTML, extracts its single JavaScript block, suppresses only the terminal UI bootstrap, and evaluates the model in Node.js with a small DOM-control adapter. Numerical inputs are obtained from the HTML defaults and the application’s `initSelects()` and `getInput()`. The model equations are not reimplemented in the experiment driver. Target-to-map synchronization is mirrored explicitly, including its rounding rules, and can be checked against the browser.

The evaluated HTML has SHA-256

```
7d0eb4bb2ba521e31151bf0ef73a958158da
bcee13bd6df2331ac5f389dd5c57
```

and the recorded runtime is Node.js v22.22.1. The source hash, baseline inputs, and scenario outputs are stored in `results/experiments.json`; scenario overrides are specified in the experiment driver. The checks cover 41 identity cases and all $41^2 = 1,681$ selectable source–target pairs. They require finite outputs at the baseline design inputs and yields within $[0, 1]$, verify expected directions for area and defect sensitivity, reproduce a fixed-seed map, check a stack-cost identity, and confirm the zero-die edge case. These checks test arithmetic consistency and execution, not empirical prediction accuracy or exhaustive invalid-input handling.

6.2 Default design and node sweep

The baseline is TSMC N12FFC with 400 mm^2 of logic, 10 W of logic power, and a 1 GHz reference frequency. SRAM capacity, SRAM power, fixed area, and NRE are zero. Wafer diameter is 300 mm, line yield is 0.95, $\alpha = 3$, package/test is \$2 per good die, volume is 10^6 , and all scenario multipliers are one. The default target is TSMC N6 and the cost basis is `intro`.

The N12-to-N6 logic-density ratio is $7.08/2.40 = 2.95$, producing 135.5932 mm^2 . The frequency ratio is $1.98/1.65 = 1.20$. Same-frequency power is 5.2041 W; at the modeled target frequency it is 6.1782 W. The defect yield is 0.83188, which becomes 0.79029 after line yield. The gross-die approximation is 464.0763, and the per-wafer assumption is \$12,493.50. Equation 11 gives \$34.0651 silicon cost and \$36.0651 after package/test.

Table 2 shows that shrinking logic is not sufficient to make cost monotonic. Under these assumptions, N7 and N6 cost \$35.96 and \$36.07 despite the N6 area improvement. N3 is the

Table 2: TSMC sweep for the fixed N12 reference design. Every value is an output of the repository model under the default introductory-price basis. Power is evaluated at the estimated target frequency.

Label (nm)	Area (mm ²)	Frequency (GHz)	Power (W)	Yield (%)	Cost (\$)
28	960.00	0.606	18.183	47.95	142.05
16	480.00	0.909	10.780	60.86	82.27
12	400.00	1.000	10.000	63.01	84.96
10	320.00	1.091	9.516	64.35	72.99
7	160.00	1.182	6.132	75.41	35.96
6	135.59	1.200	6.178	79.03	36.07
5	88.89	1.358	4.866	81.98	27.29
4	83.84	1.509	4.230	83.32	28.69
3	55.56	1.630	4.046	85.18	21.97
2	46.29	1.921	3.047	85.18	25.62

lowest-cost sampled point at \$21.97; N2 reduces area further but costs \$25.62. These are conditional results for the embedded baselines. They neither rank commercial offerings nor account for reticle feasibility: the model even permits the 960 mm² 28 nm result without a reticle-limit check.

6.3 Memory and fixed-area effects

A second scenario retains the same logic input and adds 256 Mbit SRAM, 2 W source SRAM dynamic power, and 40 mm² fixed area. Source total area becomes 461.33 mm². At N6, target area is 184.58 mm², power is 7.039 W, and cost is \$52.46. At N2 the total area is 93.01 mm², of which 46.29 mm² is logic, 6.72 mm² is SRAM, and 40 mm² is fixed. The total-area reduction from this scenario’s source is approximately 4.96 $\times$, compared with 8.64 $\times$ for its logic component.

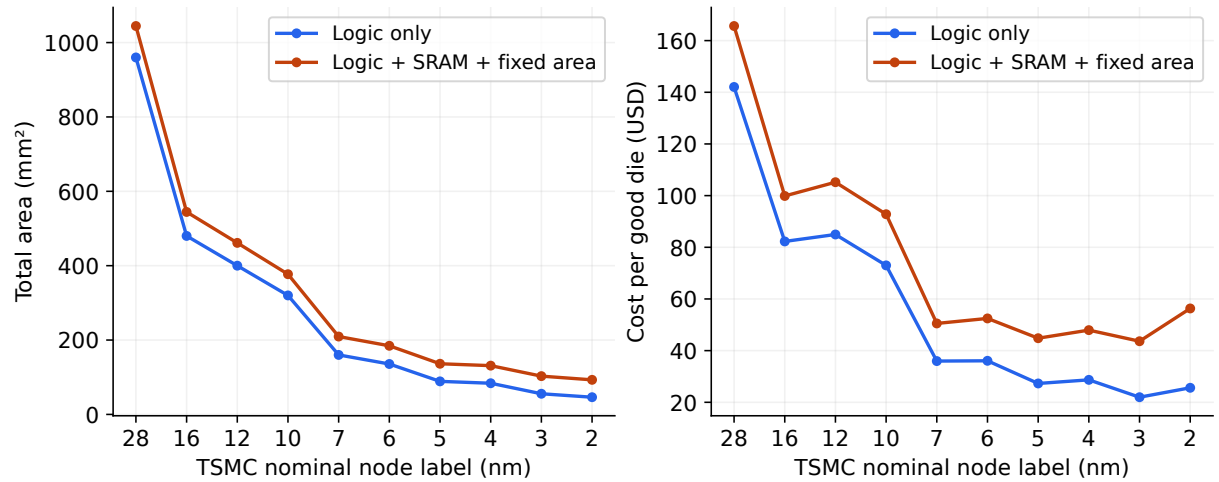


Figure 2: Effects of SRAM and fixed area on the TSMC sweep. Labels are categorical process points. Both curves use the same reference logic design and introductory-price basis. The second scenario adds 256 Mbit SRAM, 2 W source SRAM power, and 40 mm² fixed area.

At N3 this second scenario costs \$43.64, increasing to \$56.32 at N2. Figure 2 illustrates how area that scales slowly or not at all reduces the economic value of further logic shrinkage. The specific magnitudes depend on the assumed SRAM macro densities and the decision to treat fixed blocks as area-only contributions.

6.4 Cost basis and one-factor sensitivity

Changing only N6’s cost basis from introductory price (\$12,493.50 per wafer) to manufacturing cost (\$4,024) changes cost per good die from \$36.07 to \$12.97. Area, power, frequency, and yield remain identical. This difference measures a modeling convention, not an observed discount.

Table 3: One-factor changes around the default N6 case. Baseline values are 135.59 mm², 1.20 GHz, 6.178 W, 79.03%, and \$36.07.

Parameter	Value	Area	GHz	W	Yield (%)	Cost (\$)
Defect multiplier m_D	0.5	135.59	1.20	6.178	86.52	33.11
Defect multiplier m_D	2.0	135.59	1.20	6.178	66.45	42.52
Wafer multiplier m_C	0.5	135.59	1.20	6.178	79.03	19.03
Wafer multiplier m_C	2.0	135.59	1.20	6.178	79.03	70.13
Area overhead h_A	0.8	108.47	1.20	6.129	81.92	27.95
Area overhead h_A	1.2	162.71	1.20	6.225	76.27	44.86
Frequency realization h_f	0.8	135.59	0.96	5.009	79.03	36.07
Frequency realization h_f	1.2	135.59	1.44	7.347	79.03	36.07
Power guardband h_P	0.8	135.59	1.20	5.009	79.03	36.07
Power guardband h_P	1.2	135.59	1.20	7.347	79.03	36.07

Table 3 reports changes in one parameter at a time. Doubling the defect multiplier increases cost to \$42.52. A 20% logic-area overhead increases cost to \$44.86 through both fewer gross dies and lower yield. Frequency realization and the logic power guardband affect power without changing modeled yield or cost. Their equal power effects in this logic-only case follow from their product in the dynamic term; they need not be equivalent when SRAM power is present.

The structure of the equations makes local sensitivity explicit. At fixed geometry, α , line yield, wafer cost, and layer count,

$$\frac{\partial \ln C_{Si}}{\partial D_0} = \frac{LA_c}{1 + D_0 A_c / \alpha}, \quad \frac{\partial \ln C_{Si}}{\partial \ln b} = -(L - 1). \quad (15)$$

The additive package/test and NRE terms reduce the corresponding relative sensitivity of total delivered-die cost. These derivatives are properties of the implemented model, not empirically fitted elasticities.

6.5 Stacking as an assumption study

To exercise the existing W2W path, we select the repository’s Huawei LogicFold 2026 record under the same N12 reference design. This is a roadmap scenario with missing primary-source slides, not a claim about a fabricated Huawei device. The record gives 88.84 mm² footprint, 1.879 GHz, and 16.748 W, with single-layer post-line yield $q = 0.60413$ and $b = 0.995$.

At two layers, modeled stack yield is 36.31% and cost is \$96.91; at three layers they become 21.83% and \$238.84. The selected footprint, power, and frequency do not change with layer count. Removing the common \$2 package/test term gives the exact relationship

$$\frac{C_{Si,3}}{C_{Si,2}} = \frac{3}{2qb} \approx 2.495. \quad (16)$$

The additional wafer and compounded yield loss explain the increase. The experiment does not optimize layer partitioning, bonding technology, interconnect overhead, or thermal behavior.

6.6 Analytical versus spatial yield

On first load, the map has independent defaults: 20 mm square dies, $D_0 = 0.28 \text{ cm}^{-2}$, a \$4,024 wafer, 3 mm edge exclusion, PPM=1,000, radial boost $\beta = 0.6$, and seed 2026. It reports 148

full dies, 52 passing dies, and expected yield 32.70%. This initial map is not the default N6 analytical scenario.

After target synchronization the dimensions become 11.64 mm by 11.64 mm, with $D_0 = 0.14$ and wafer cost rounded to \$12,494. Retaining the map’s edge and PPM defaults gives 448 full dies, expected yield 76.7839%, and expected cost \$38.32. Seed 2026 yields 352 passing dies (78.5714%) and sampled cost \$37.49. The analytical card gives 79.0286% and \$36.07, owing to its different geometry, lack of explicit PPM/radial losses, and unrounded inputs.

Setting only map radial boost and PPM to zero raises expected yield to 79.0394%, close to the analytical 79.0286%; the remaining difference is consistent with rounded die dimensions. Expected map cost remains \$37.28 because its integer placement and explicit edge exclusion still differ from Eq. 8. Across seeds 1–100 with the synchronized default map, mean sampled yield is 76.6496%, with sample standard deviation 1.9500 percentage points and range 70.5357–81.2500%. The closeness of the sample mean to 76.7839% is a sampling sanity check, not a validation against fab data or a proof of hash independence.

7 Limitations and validation needs

The dominant limitation is parameter provenance. Several process factors are interpolated, costs and defect densities are assumed, and the uploaded materials referenced by the original interface are absent. The qualitative confidence labels cannot support a quantitative uncertainty interval. Published PPA percentages can also describe different voltage, library, utilization, or iso-power/iso-frequency conditions. Treating them as independent scalar factors can combine operating points that are not jointly achievable.

Physical-design effects are represented only by user multipliers. There is no RTL synthesis, floorplanning, placement, routing, parasitic extraction, static timing analysis, IR-drop calculation, or thermal simulation. The model omits IO/analog power, explicit interconnect power, workload activity traces, SRAM leakage, and memory redundancy. It imposes no reticle limit and does not optimize die placement. Its negative-binomial law uses total die area as effective defect-sensitive area.

W2W estimates assume equal layers and independent pass events. The model does not allocate different circuits to layers, model bonding overhead cost separately, account for die-to-wafer selection, or characterize inter-layer correlation. A layer-count increase changes cost and yield without generating a new physical design. The radial yield map likewise does not model observed spatial defect clusters.

Arithmetic checks cannot resolve these limitations. A predictive evaluation would require designs with matched source and target cell libraries, comparable physical-design constraints, characterized SRAM macros, measured or signoff-quality power and timing, and sourced yield/cost assumptions. Calibration designs should be separated from held-out designs. Such a study could report area, frequency, and power errors individually, and propagate yield/cost parameter uncertainty. This paper contains no such dataset, and leaves that evaluation to future work.

8 Conclusion

Silicon Analysis provides a compact, inspectable environment for exploring the relationships between process factors, design composition, yield, and cost. Its useful distinction is the separation of logic, SRAM, and fixed area, together with analytical and spatial views of manufacturing assumptions. The reproduced experiments show how cost-basis selection, unscaled area, and stacked yield can dominate conclusions drawn from logic density alone. The artifact supports transparent scenario analysis; its predictive use requires calibration and independent validation beyond the repository’s default library.

Artifact availability and acknowledgments

The repository is distributed under Apache-2.0. This manuscript is accompanied by its LaTeX source, vector figures, editable draw.io architecture, numerical results, and reproduction instructions. This documentation and manuscript draft were prepared with AI assistance. Manuscript content requires author review before submission.

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